

Data Wrangling 🤠

Programs

Lesson 5

Programs

- Programs are sequences of instructions that we tell a computer to execute one-by-one
- Unlike code we write in the REPL, we can save the contents of a program to run at a later time

Demo: Graduation Year

```
# grad_year.py  
year_born = 2026  
school_years = 18  
  
grad_year = year_born + school_years
```

Run the program with the command: `python -m lessons.grad_year`

Demo: Graduation Year

```
# grad_year.py
year_born = 2024
school_years = 18

grad_year = year_born + school_years

print(f"Someone born in {year_born} will graduate in {grad_year}!")
```

Run the program with the command: `python -m lessons.grad_year`

Memory in a Program

- When you run code, your computer sets aside a chunk of memory to keep track of what is happening
- Memory is a snapshot of the state of the program

Demo: Variables Revisited

```
# variables.py
x = 30
y = 10
z = 5

added = x + y + z
x = y
compared = x == y
x = x + 1
compared = x == y

print(f"x, y, z: {x}, {y}, {z}")
print(f"added_up: {added}")
print(f"compared: {compared}")
```